

7 SEM

1.

```
df_raw <- read.csv("data.csv")
df <- na.omit(df_raw)
head(df)
```

```
##   subj good_bad goodness q1 q2 q3 q4 q5 q6 q7 q8 q9 q10 q11 q12 q13 q14 q15
## 1    1         0      -3  2  2 -2 -1  1  1 -2 -2 -1  -2  -2  -1  -2  -2  -1
## 2    1         1       3 -2 -2  2  2 -2 -2  1  2  2  2  2  2  1  2  2
## 3    2         0      -2  1  2 -1 -2  2  1 -1 -2 -2  -1  -1  -2  -1  -1  -2
## 4    2         1       2 -1 -1  1  1 -1 -2  1  1  2  1  1  1  2  1  2
## 5    3         0      -3  2  2 -2 -2  2  2 -2 -2 -2  -2  -2  -2  -2  -2  -2
## 6    3         1       3 -2 -2  2  2 -2 -1  2  2  2  2  2  2  2  2  2
##   Gender Age
## 1      M  21
## 2      M  21
## 3      F  20
## 4      F  20
## 5      M  22
## 6      M  22
```

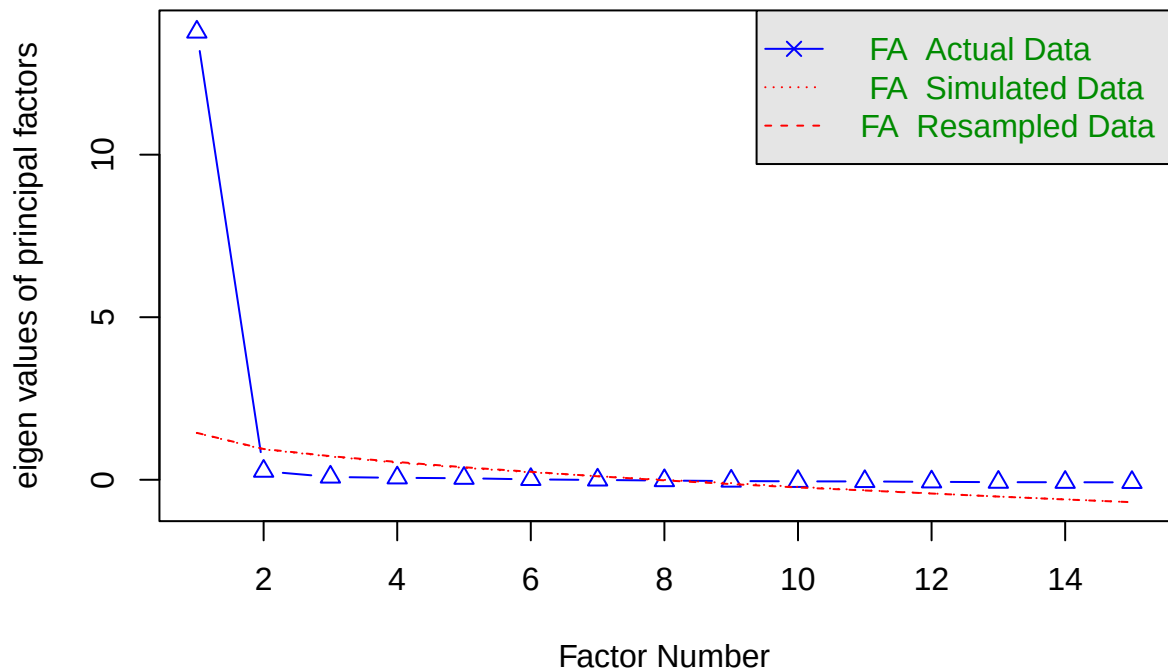
```
cat("      (N):", nrow(df))
```

```
##      (N): 42
```

1:

```
df_fa <- df[, 4:18]
fa.parallel(df_fa, fa = "fa", n.iter = 100)
```

Parallel Analysis Scree Plots



Parallel analysis suggests that the number of factors = 1 and the number of components = NA

2 & 3:

```
# =3,
fa_result <- factanal(df_fa, factors = 3, rotation = "promax")
print(fa_result, cutoff = 0.3)
```

```
##
## Call:
## factanal(x = df_fa, factors = 3, rotation = "promax")
##
## Uniquenesses:
##   q1   q2   q3   q4   q5   q6   q7   q8   q9  q10  q11  q12  q13
## 0.055 0.074 0.057 0.016 0.048 0.091 0.093 0.035 0.028 0.005 0.056 0.043 0.037
##   q14  q15
## 0.033 0.078
##
## Loadings:
##   Factor1 Factor2 Factor3
## q1 -0.642 -0.319
## q2 -0.732
## q3  0.802
## q4      0.879
## q5 -0.768
## q6 -0.440 -0.565
```

```
## q7    0.449    0.443
## q8          0.730
## q9    0.935
## q10         0.830
## q11    0.529    0.441
## q12    0.328    0.723
## q13    0.692          0.324
## q14    0.578          0.348
## q15    0.479    0.449
##
##              Factor1 Factor2 Factor3
## SS loadings      4.914   3.677   0.388
## Proportion Var    0.328   0.245   0.026
## Cumulative Var    0.328   0.573   0.599
##
## Factor Correlations:
##              Factor1 Factor2 Factor3
## Factor1      1.000   0.847   0.627
## Factor2      0.847   1.000   0.615
## Factor3      0.627   0.615   1.000
##
## Test of the hypothesis that 3 factors are sufficient.
## The chi square statistic is 141.29 on 63 degrees of freedom.
## The p-value is 6.11e-08
```

SEM

```
model <- '
#
Strategy =~ q7 + q8 + q9 + q10
Unity    =~ q11 + q12 + q13
Barrier  =~ q1 + q2 + q5 + q6

#
goodness ~ Strategy + Unity + Barrier
'

fit <- sem(model, data = df)
```

4: SEM

P

```
#              (std.all)
params <- parameterEstimates(fit, standardized=TRUE)

# ~
reg_results <- subset(params, op == "~", select = c("lhs", "rhs", "std.all", "pvalue"))

#
colnames(reg_results) <- c(" ", " ", " ", " ", "p ")
print(reg_results)
```

```
##          P
## 12 goodness Strategy    -0.562 0.827
## 13 goodness  Unity     -0.054 0.940
## 14 goodness Barrier    -1.608 0.625
```

5: SEM

CFI, RMSEA

```
#
fitMeasures(fit, c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "srmr"))

##   chisq    df pvalue    cfi    tli  rmsea  srmr
## 166.251 49.000  0.000  0.907  0.875  0.239  0.020
```

6: SEM

```
semPaths(fit,
  whatLabels = "std", #
  layout = "tree", #
  edge.label.cex = 1, #
  curvePivot = TRUE, #
  sizeMan = 5, #
  sizeLat = 8) #
```

